

TECHNICAL DATA INSPECTION NOTEBOOK

HealthConnect Experience Lab — Data Analytics Track | Week 4

Supporting Technical Evidence — Notebook Design & Inspection Framework

NOTEBOOK POSITIONING

The Technical Data Inspection Notebook is the technical evidence layer of the hybrid documentation approach. It does not replace the Initial Analysis Document; it makes the technical observations used to frame the Week 4 analysis reproducible, traceable, and verifiable.

1. Notebook Objective

The objective is to technically inspect the HealthConnect Appointment Dataset, verify its alignment with the Data Dictionary, identify and document the main data quality issues, and assess whether the data are sufficiently ready for the analytical work planned in subsequent project stages.

The notebook is intentionally limited to inspection, validation, and traceability. It should not become a full EDA that already answers the Business Questions, nor should it generate premature business recommendations.

2. Documentation Principle

Each technical check will systematically follow the sequence below:

Purpose → Code → Result → Interpretation

This structure enables the mentor or any technical reviewer to understand why a check is performed, how it is executed, what result it produces, and which analytical implication should be retained.

3. Proposed Final Structure

#	SECTION	PURPOSE
1	Notebook Context & Objective	Present the HealthConnect context, notebook role, and Week 4 analytical boundary.
2	Environment Setup & Data Loading	Import the required libraries and load the dataset and Data Dictionary.
3	Data Dictionary Alignment	Verify correspondence between documented variables and observed dataset columns.
4	Dataset Structure & Grain Validation	Confirm dimensions, columns, analytical grain, and identifiers.
5	Data Type Assessment	Compare observed and expected data types and identify required conversions.
6	Missing Values Assessment	Quantify and qualify missingness, including structural missingness where relevant.
7	Duplicate & Identifier Checks	Check exact duplicates, appointment_id uniqueness, and repeated patient_id values.
8	Categorical & Numerical Validity Checks	Inspect categories, value ranges, and obvious invalid values.
9	Cross-Field Consistency Checks	Test consistency rules across related variables.
10	Repeated patient_id Consistency Checks	Assess consistency of attributes linked to repeated patients.
11	Data Quality Issue Register	Consolidate issues, evidence, risk level, and proposed handling.
12	Analytical Readiness Assessment	Determine what is usable, conditional, or unsuitable for future analysis.
13	Notebook Conclusion	Summarize inspection findings and prepare the transition to future analysis.

4. Expected Technical Checks

- Reproducible loading of HealthConnect_Appointment_Data.csv and HealthConnect_Data_Dictionary.xlsx.
- Display of dataset dimensions, column names, and a controlled preview of the first records.
- Validation of the analytical grain: one row should represent one appointment record.
- Check of appointment_id uniqueness and assessment of repeated patient_id values.
- Comparison of observed data types with the definitions provided in the Data Dictionary.

- Count and percentage of missing values by variable, distinguishing data-quality issues from structural missingness where justified.
- Search for exact duplicates and identifier inconsistencies.
- Review of categorical modalities and checks for clearly invalid numerical values or ranges.
- Cross-field consistency checks where multiple variables describe the same operational or historical context.
- Specific consistency checks for attributes associated with repeated patient_id values to prevent unreliable longitudinal reconstruction.
- Consolidation of technical findings in a Data Quality Issue Register.

5. Data Quality Issue Register — Proposed Structure

Issue	Evidence	Risk	Proposed Treatment	Analytical Impact
To be documented after execution	Code + result	Low / Medium / High	Keep / Correct / Flag / Exclude	Impact on future analysis
Example: cross-field inconsistency	Reproducible check	To assess	Decide based on context	Affected population, metric, or interpretation

6. What the Notebook Should Not Do in Week 4

- Calculate and interpret final answers to the Business Questions.
- Fully test analytical hypotheses H1–H12.
- Build an EDA consisting of exploratory charts unrelated to technical inspection.
- Produce a final dashboard.
- Issue final business recommendations.
- Turn an observed association into a causal conclusion.

7. Interpretation & Traceability Rules

No material result should be manually entered as a final technical finding. Key observations will be recalculated from the raw source files so that the notebook functions as genuine reproducible evidence.

Notebook results must remain consistent with the Initial Analysis Document. If a technical check changes a previously documented observation, the main document must be corrected before submission to preserve cross-deliverable consistency.

8. Expected Outcome at Notebook Completion

At the end of the technical inspection, the notebook should enable four questions to be answered clearly: (1) What is the actual structure of the dataset? (2) Which data quality issues were identified? (3) Which variables or analytical populations require specific precautions? (4) Are the data sufficiently ready to proceed to Business Question analysis in the next phase?

Technical Data Inspection Notebook = Supporting Technical Evidence

Technical inspection, validation, traceability, and preparation — without premature EDA.

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